DATA SCIENCE & ANALYTICS · SESSION 07

Insights Report & Dashboard

Use Case 1, part two: turning the analysis into an interactive story and a memo

Prof.Dr.Chaiwat N. · 2026

2026 · Bachelor Program · 3 hrs · Workshop Format · Hands-on Coding

Session Agenda

01 **From Analysis to Insight**
What a report is — and isn't

02 **Interactive Charts with Plotly**
px.bar, px.line, px.scatter, sunburst

03 **Assembling a Dashboard**
A multi-panel view in one notebook

04 **The Insight Memo**
Three points and a recommendation

05 **Presenting to Non-Technical Readers**
Translating numbers into decisions

06 **Workshop Block**
Write the 3-point memo — graded

01

PART 01

From Analysis to Insight

An analysis is a pile of correct numbers. An insight report selects the few that matter and tells the reader what to do about them.

A report is not your notebook — it's the three things the reader must know

What a report is

Not a data dump

Nobody acts on twenty charts. A report ruthlessly selects the few findings that change a decision.

Structured

Executive summary first, then key findings, then recommendation. Busy readers get the answer in the first lines.

Answers 'so what'

Every number is paired with its implication. A figure without a 'therefore' doesn't belong in the report.

Ends in action

The whole document exists to support one recommendation. If it doesn't lead somewhere, cut it.

Source: DS Course, Session 7, 2026

02

PART 02

Interactive Charts with Plotly

Plotly Express makes charts the reader can hover, zoom, and filter — the same grammar as before, now interactive and presentation-ready.

Plotly Express: interactive charts in one line

Hover, zoom, explore

Same grammar

px.bar, px.line, px.scatter mirror what you know — but the output is interactive and shareable as HTML.

Hover detail

Readers see exact values on hover, so you don't clutter the chart with labels. The detail is on demand.

When to use

For dashboards and reports people explore themselves. Static Matplotlib is still better for print and slides.

PYTHON · PLOTLY

```
import plotly.express as px

fig = px.bar(rev_by_cat, x='category',
             y='revenue', color='year',
             barmode='group',
             title='Revenue by Category')
fig.show()

# trend line, one per category
px.line(monthly, x='month', y='revenue',
        color='category')
```

Same chart grammar — now interactive.

A sunburst and a scatter: showing hierarchy and relationship

Two views that earn their place

Sunburst

Revenue broken down category → product as nested rings. Shows the contribution hierarchy in one glance.

Bubble scatter

Revenue vs margin, bubble size = volume. Three variables at once — find the high-revenue, high-margin winners.

Restraint

Interactive doesn't mean busy. Each chart still answers one question; the interactivity just adds detail.

PYTHON · HIERARCHY

```
px.sunburst(df, path=['category','product'],
            values='revenue')

px.scatter(prod, x='margin', y='revenue',
           size='quantity', color='category',
           hover_name='product')
```

Pick the chart that answers the question — not the flashiest.

03

PART 03

Assembling a Dashboard

Bring the key charts into one coherent panel the reader can scan top to bottom — and export it as a standalone HTML file.

A dashboard is a curated panel, not every chart you made

One screen, four views

Curate

Choose the four charts that together answer the brief. A dashboard with everything answers nothing.

Arrange

Most important top-left — that's where eyes land first. Flow from headline KPI to supporting detail.

Consistent

Shared colours, aligned titles, same number formats. Consistency signals care and aids comparison.

Export

Write it to a single HTML file anyone can open in a browser — no Python, no setup. The shareable artifact.

Source: DS Course, Session 7, 2026

04

PART 04

The Insight Memo

Three findings, each with evidence and an implication, ending in one clear recommendation — the one-page document a manager actually reads.

The 3-point memo forces clarity: three findings, one recommendation

Structure the message

Finding	A specific, evidenced statement: 'Electronics revenue grew 24% while margin fell 6 points.' Number plus direction.
Implication	Why it matters: 'Growth is being bought with discounts that erode profit.' The bridge to a decision.
Recommendation	One concrete action: 'Cap Electronics discounting at 15% and protect margin.' Specific and assignable.
Three, not ten	Three points fit in a reader's head and force you to prioritize. The discipline is the value.

Source: DS Course, Session 7, 2026

05

PART 05

Presenting to Non-Technical Readers

Your audience cares about the decision, not your code. Translate every technical result into plain business language.

Translate: the reader wants the decision, not the method

Speak business

Drop the jargon

'Revenue concentration' not 'Pareto index'. If a word needs a definition, replace it with the plain version.

Lead with the answer

Conclusion first, evidence second. Don't make a busy reader follow your derivation to reach the point.

Numbers in context

'฿2.4M' means little; '฿2.4M — up 24% and our biggest gain' means something. Always anchor a number.

Anticipate the question

End each point by answering the obvious 'so what should we do?' before they have to ask.

Source: DS Course, Session 7, 2026

06

PART 06

Workshop Block

Write a one-page, 3-point insight memo from the Use Case 1 analysis, ending in a single recommendation — then swap for peer review.

By the end of this block, every student has a one-page memo a manager could act on

Workshop · 30 Minutes + Peer Review

Pick three findings	From all of Session 6–7, choose the three that most change the decision. Kill your darlings.
Write each as F-I-R	Finding, implication, recommendation. One tight paragraph each — evidence included, jargon removed.
Add supporting charts	One Plotly chart per point, titled with its finding. The memo plus charts is your deliverable.
Peer review	Swap memos. Can your partner state your recommendation in one sentence? If not, it isn't clear enough yet.

Source: Workshop, Session 7, 2026

KEY TAKEAWAY

Analysis informs. A clear recommendation is what gets acted on.